Jaehyeon Seo

Department of Mathematics, Yonsei University
jaehyeonseo@yonsei.ac.kr jaehyeonseo.com
Google Scholar arXiv GitHub LinkedIn

Last updated: November 2026

Research Interests

Extremal and and probabilistic combinatorics, graph homomorphism inequalities, independent sets and colorings

Education

Combined Master's and Doctoral Program in Mathematics Mar 2023–Present
Yonsei University Seoul, Korea
- Supervised by Joonkyung Lee

Graduate Study in Mathematics (not completed) Mar 2020–Aug 2022
Korea Advanced Institute of Science and Technology Daejeon, Korea
- Supervised by Jaehoon Kim

Bachelor's in Mathematics Feb 2016–Feb 2020
Pohang University of Science and Technology Pohang, Korea
- Magna Cum Laude

High school for science-gifted students Feb 2013–Feb 2016
Korea Science Academy of KAIST Busan, Korea

Publications and Preprints

Publications

3. **Hamilton transversals in tournaments**. *Combinatorica* **44** (2024), 1381–1400. [arxiv] [journal]
   With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee.
2. **On a rainbow extremal problem for color-critical graphs**. *Random Structures & Algorithms* **64** (2024), 460–489. [arxiv] [journal]
   With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee, Hong Liu.
1. **On 1-Subdivisions of Transitive Tournaments**. *The Electronic Journal of Combinatorics* **29**(1) (2022), P1.51. [arxiv] [journal]
   With Jaehoon Kim, Hyunwoo Lee.

Preprints and Manuscripts

7. **Fractional colorings of K_r-free graphs with local demands**. Submitted (2026). [arxiv]
   With Ewan Davies, Abhishek Dhawan, Abhishek Methuku, Huy Nguyen.
6. **Antiferromagnetic models are clique-minimizing**. Submitted (2026). [arxiv]
   With Joonkyung Lee.
5. **Degree-sequence bounds for independent sets via multivariate local occupancy**. Preprint (2026). [arxiv]
   With Ewan Davies, Juspreet Singh Sandhu, Brian Tan.
4. **Lower bounds for multivariate independence polynomials and their generalisations**. Submitted (2026). [arxiv]
   With Joonkyung Lee.
3. **Semi-Autonomous Mathematics Discovery with Gemini: A Case Study on the Erdős Problems**. Preprint (2026). [arxiv]
   With Tony Feng, Trieu Trinh, Garrett Bingham, Jiwon Kang, Shengtong Zhang, Sang-hyun Kim, Kevin Barreto, Carl Schildkraut, Junehyuk Jung, Carlo Pagano, Yuri Chervonyi, Dawsen Hwang, Kaiying Hou, Sergei Gukov, Cheng-Chiang Tsai, Hyunwoo Choi, Youngbeom Jin, Wei-Yuan Li, Hao-An Wu, Ruey-An Shiu, Yu-Sheng Shih, Quoc V. Le, Thang Luong.
2. **Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials**. Submitted (2025). Extended abstract accepted to FPSAC 2025. [arxiv]
   With Joonkyung Lee, Jaeseong Oh.
1. **Transversal cycles and paths in tournaments**. Submitted (2024). [arxiv]
   With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee.

Research Experience and Visits

Research Visitor — May 24–31, 2026
Colorado State University (Host: Ewan Davies) — Fort Collins, CO, US

Visiting Student Researcher — Jan 2026–Jul 2026
California Institute of Technology (Host: David Conlon) — Pasadena, CA, US

Research Visitor — Jun 9–14, 2025
Shanghai Center for Mathematical Sciences (Host: Yaobin Chen) — Shanghai, China

Researcher — Oct 2022–Feb 2023
Hanyang University Industry–University Cooperation Foundation — Seoul, Korea

Visiting Student — Apr 2022–Aug 2022
Extremal Combinatorics and Probability Group, Institute for Basic Science — Daejeon, Korea

Exchange Student — Feb 2018–Jul 2018
Graz University of Technology — Graz, Austria

Talks

- **2026 KMS Annual Meeting**, GIST — Gwangju, Korea
  Antiferromagnetic models are clique-minimizing — Oct 29, 2026
- **2026 Combinatorics Workshop**, Konjiam Resort — Gwangju, Gyeonggi, Korea
  Fractional colorings of K_r-free graphs with local demands — Oct 20, 2026
- **ASIACOMB 2026**, Daejeon Convention Center — Daejeon, Korea
  Degree-sequence bounds for independent sets via multivariate local occupancy — Aug 24, 2026
- **Workshop for Young Mathematicians in Korea 2026**, KAIST — Daejeon, Korea
  What counts do cliques minimize? Independent Sets, colorings, and more (invited) — Aug 20, 2026
- **Combinatorics at the Confluence**, Carnegie Mellon University — Pittsburgh, PA, US
  Clique-minimizing models (poster) — Jul 20, 2026
- **2026 SIAM Conference on Discrete Mathematics**, UCSD — San Diego, CA, US
  Lower bounds for multivariate independence polynomials and their generalisations — Jun 23, 2026
- **Flags in the Mountains**, CSU Mountain Campus — Bellvue, CO, US
  Clique-minimizing models (lightning talk) — Jun 4, 2026
- **Graduate Student Combinatorics Conference 2026**, UIC — Chicago, IL, US
  Lower bounds for multivariate independence polynomials and their generalizations — Mar 28, 2026
- **Math Graduate Student Seminar**, Caltech — Pasadena, CA, US
  Extremal questions on independent sets, colorings, and graph homomorphisms (invited) — Feb 24, 2026
- **RS&A 2025**, TU Wien — Vienna, Austria
  Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials — Aug 5, 2025
- **FPSAC 2025**, Hokkaido University — Sapporo, Japan
  Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials (poster) — Jul 21, 2025
- **Combinatorics Student Seminar**, Shanghai Center for Mathematical Science — Shanghai, China
  Graph homomorphism inequalities and Lorentzian polynomials (invited) — Jun 11, 2025
- **CanaDAM 2025**, University of Ottawa — Ottawa, Canada
  Graph homomorphism inequalities and Lorentzian polynomials (invited) — May 21, 2025
- **4th East Asia Workshop on Extremal and Structural Graph Theory**, Sun Yat-Sen University — Guangzhou, China
  Graph homomorphism inequalities and Lorentzian polynomials — Mar 30, 2025
- **Sinchon Seminar of Graduates 5**, Ewha Womans University — Seoul, Korea
  Graph homomorphisms and Lorentzian polynomials (invited) — Dec 27, 2024
- **KIAS Graduate Student Meeting in Pure Mathematics**, KIAS — Seoul, Korea
  Graph homomorphisms and Lorentzian polynomials — Oct 21, 2024

- **2024 Combinatorics Workshop**, Chungbuk National University — Cheongju, Korea
  Hamilton transversals in tournaments — Aug 29, 2024
- **2024 Korea Student Combinatorics Workshop**, Gongju Hanok Village — Gongju, Korea
  Extremal graph theory and rainbow substructures — Jul 30, 2024
- **2023 KMS Spring Meeting**, Daejeon Convention Center — Daejeon, Korea
  Rainbow oriented Hamiltonian paths and cycles in tournaments — Apr 29, 2023
- **KAIST Math Graduate Student Seminar**, KAIST — Online
  How can we find a rainbow subgraph? (invited) — Mar 24, 2022
- **Discrete Math Seminar**, IBS DIMAG — Daejeon, Korea
  A rainbow Turán problem for color-critical graphs (invited) — Jan 18, 2022
- **2021 Combinatorics Workshop**, The Bloomvista — Yangpyeong, Korea
  On multicolor extremal problems — Dec 21, 2021

Awards and Grants

- BK21FOUR International Joint Training Support of Outstanding Graduate Students — Jan 2026–Jul 2026
  (4단계 BK21사업 우수대학원생 국제공동연수)
- Best Student from the Brain Korea (BK) Evaluation Department of Mathematics, Yonsei University — Spring 2024/2025/2026
- Excellence Award Spring Poster Presentation, School of Mathematics and Computing, Yonsei University — Spring 2024
- Dean's List POSTECH Department of Mathematics — Fall 2016, Spring/Fall 2017, Spring 2019
- Excellence Prize POSTECH Programming Contest — 2016
- National Scholarship (Science and Engineering) Korea Student Aid Foundation — Feb 2016–Feb 2020

Teaching Experience

Teaching Assistant — Mar 2023–Dec 2025
Yonsei University — Seoul, Korea
- Combinatorics — Fall 2025; Fall 2024
- Discrete Mathematics — Spring 2025; Spring 2024
- Linear Algebra (2) — Fall 2023
- Mathematics and Programming — Spring 2023

Teaching Assistant — Mar 2020–Aug 2022
Korea Advanced Institute of Science and Technology — Daejeon, Korea
- Probability and Statistics, Scientific Writing (for math majors) — Spring 2022
- Calculus 1, Introduction to Graph Theory — Fall 2021
- Introduction to Linear Algebra, Introduction to Number Theory — Spring 2021
- Differential Equations and Applications, Introduction to Differential Geometry — Fall 2020
- Differential Equations and Applications — Spring 2020

Excellent Student Teaching Assistant — Mar 2019–Dec 2019
Pohang University of Science and Technology — Pohang, Korea
- Calculus 2 — Fall 2019
- Calculus 1 — Spring 2019

Teaching Assistant — Oct 2016–Feb 2017
KAIST Cyber Gifted Education — Online
- Introduction to Number Theory, Linear Algebra — Fall 2016

Outreach

Teaching Assistant — Jan 2018
POSTECH Potential Development Camp for High School Students — Pohang, Korea
(POSTECH 잠재력개발과정)
- Tutoring high school students in undergraduate-level mathematics

Academic Service

- Co-organizer, Korean Student Combinatorics Workshop 2026 Winter — Feb 2–6, 2026
- Co-organizer, Korean Student Combinatorics Workshop 2025 — Aug 20–24, 2025
- Co-organizer, Discrete Analysis Seminar — Apr 2023–Present

Peer Review Activities

- SIAM Journal on Discrete Mathematics — 2026
- Combinatorics, Probability and Computing; SIAM Journal on Discrete Mathematics — 2025
- Journal of Graph Theory — 2024

Student Seminars

- Co-organizer, Discrete Analysis & Insights Seminar at Yonsei — Oct 2024–Present
  Student-led seminar at Yonsei University, primarily focusing on extremal combinatorics
- Participant, Deep Seminar in Combinatorics — Jan 2024–Sep 2025
  Student-led seminar at KAIST
- Participant, Seminars on Gowers Norms — Jul 2025–Nov 2025

Selected Academic Events

Outside of Korea

- **ICM 2026** International Congress of Mathematicians, Pennsylvania Convention Center — Philadelphia, PA, US. Jul 23–30, 2026
- **Combinatorics at the Confluence**, Carnegie Mellon University — Pittsburgh, PA, US. Jul 20–22, 2026
- **2026 SIAM Conference on Discrete Mathematics**, UCSD — San Diego, CA, US. Jun 22–25, 2026
- **Flags in the Mountains**, Colorado State University Mountain Campus — Bellvue, CO, US. Jun 1–5, 2026
- **Graduate Student Combinatorics Conference 2026**, UIC — Chicago, IL, US. Mar 27–29, 2026
- **RS&A 2025** Random Structures & Algorithms, TU Wien — Vienna, Austria. Aug 4–8, 2025
- **FPSAC 2025** Formal Power Series and Algebraic Combinatorics, Hokkaido University — Sapporo, Japan. Jul 21–25, 2025
- **CanaDAM 2025** Canadian Discrete and Algorithmic Mathematics Conference, University of Ottawa — Ottawa, Canada. May 20–23, 2025
- **4th East Asia Workshop on Extremal and Structural Graph Theory**, Sun-Yat Sen University — Guangzhou, China. Mar 27–31, 2025
- **Summit280**, Eötvös University — Budapest, Hungary. Jul 8–12, 2024
- **BCC 2024** British Combinatorial Conference, Queen Mary University of London — London, UK. Jul 1–5, 2024
- **3rd East Asia Workshop on Extremal and Structural Graph Theory**, The Southern Beach Hotel & Resort Okinawa — Okinawa, Japan. Nov 1–5, 2023

AI and Mathematics

- **AI and Mathematics**, KIAS — Seoul, Korea. Aug 3–7, 2026
  Invited to the fourth week on "Algebra, Combinatorics, and Topology"
- **Summer School on Formalization in Lean**, ICARM — Pittsburgh, PA, US. Jul 7–18, 2026
- **FrontierMath: Open Problems workshop**, USC Hotel — Los Angeles, CA, US. Jun 9, 2026
- **2025 KIAS Winter School on Mathematics and AI**, Park Roche — Jeongseon, Korea. Dec 2–5, 2025
  Attended team sessions on ML and Discrete Mathematics, focusing on FunSearch
- **Machine Learning and Mathematics**, KIAS — Seoul, Korea. Jul 7–11, 2025
- **AI for Mathematics Workshop on Formalization**, KIAS — Seoul, Korea. Dec 17, 2024
- **Samsung Global Research Symposium: Mathematical Theory of AI**, InterContinental Seoul COEX — Seoul, Korea. Aug 1–3, 2023

International Events in Korea

- **ASIACOMB 2026**, Daejeon Convention Center — Daejeon, Korea. Aug 24–28, 2026
- **5th East Asia Workshop on Extremal and Structural Graph Theory**, Fraser Place Namdaemun — Seoul, Korea. Nov 27–30, 2025
- **1st Early Career Workshop in Extremal Combinatorics**, IBS ECOPRO — Daejeon, Korea. Oct 20–30, 2025
- **KIAS-KAIST Workshop on current challenges in mathematics**, KIAS — Seoul, Korea. Aug 25–28, 2025
- **Discrete Analysis at Yonsei**, Yonsei University — Seoul, Korea. Jun 16–20, 2025
- **2024 IBS-DIMAG Workshop on Combinatorics and Geometric Measure Theory**, IBS — Daejeon, Korea. Jul 14–19, 2024

Schools

- **2026 Summer School on Combinatorics and Algorithms**, KAIST — Daejeon, Korea. Aug 10–14, 2026
  Daniel Dadush, Algorithms & Geometry of Linear Programming
  Magnus Wahlström, Matroids, delta-matroids, and applications
- **IBS-DIMAG Winter School on Graph Minors**, IBS DIMAG — Daejeon, Korea. Feb 3–19, 2025
  Sebastian Wiederrecht and Maximilian Gorsky, Graph Minor Structure Theorem
- **ECOPRO–CAS–SDU Joint Winter Course**, IBS ECOPRO — Daejeon, Korea. Jan 6–10, 2025
  Huy Tuan Pham, Recent developments in probabilistic combinatorics: New synergies
- **IBS–SDU Frontiers of Combinatorics, Part 2**, IBS ECOPRO — Daejeon, Korea. Aug 5–9, 2024
  Tung Nguyen, Recent work on the Erdös–Hajnal conjecture
- **2024 Summer School on Combinatorics and Algorithms**, KAIST — Daejeon, Korea. Jul 22–26, 2024
  Chien-Chung Huang, Combinatorial Optimization
  Sebastian Wiederrecht, From treewidth to grid minor theorem
- **IBS–CAS–SDU Joint Course**, IBS ECOPRO — Daejeon, Korea. Feb 26–Mar 1, 2024
  Bjarne Schülke, Hypergraph Turán problems